DBMS Normalization Cheat Sheet (1NF to BCNF)

Normalization is a database design technique used to reduce redundancy and improve data integrity by organizing attributes and relations based on functional dependencies.

Key Terms

- **Attribute:** A column in a table
- **Tuple:** A row in a table
- **Primary Key:** Uniquely identifies a tuple
- **Candidate Key:** Minimal set of attributes that can act as a primary key
- **Functional Dependency (FD):** $X \rightarrow Y$ means X determines Y

First Normal Form (1NF)

- All attributes contain **atomic (indivisible)** values
- No repeating groups or multi-valued attributes
- Each record can be uniquely identified
- **Example Fix:** Split multi-valued attributes into separate rows

Second Normal Form (2NF)

A table is in 2NF if it is in 1NF and:

- No **partial dependency**
- All non-prime attributes are fully functionally dependent on the whole primary key
- Applies only when the primary key is **composite**
- **Example Fix:** Move attributes depending on part of the key to a new table

Third Normal Form (3NF)

A table is in 3NF if it is in 2NF and:

- No **transitive dependency**
- Non-prime attributes do not depend on other non-prime attributes
- For every FD $X \rightarrow A$, either X is a super key or A is a prime attribute
- **Example Fix:** Separate dependent attributes into another table

Boyce–Codd Normal Form (BCNF)

A stronger version of 3NF. A table is in BCNF if:

- For every functional dependency $X \rightarrow Y$, **X must be a super key**
- Eliminates anomalies that may still exist in 3NF
- May result in loss of dependency preservation (but not loss of data)
- **Example Fix:** Decompose tables so determinants are super keys

Quick Comparison

1NF: Atomic values, no repeating groups
2NF: 1NF + no partial dependency
3NF: 2NF + no transitive dependency
BCNF: Every determinant is a super key

Exam Tips (C-CAT Friendly)

- Always find the **candidate keys** first
- Check **partial dependency** only for composite keys (2NF)
- Check **transitive dependency** for 3NF
- If determinant is not a super key $\rightarrow$ violates BCNF